FINAL DATA ANALYTICS PROJECT

Global Air Quality Analytics Case Study

Problem statement, data preparation, analysis, Power BI dashboard and key business insights

23,463	72.01	55	81.7%	68.52
records analysed	average AQI	median AQI	Good or Moderate	PM2.5 average

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Internship period: 6 October to 5 November 2026

SUBMITTED
October 2026

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Project Overview

This project analyses the Global Air Pollution Dataset from Kaggle and turns it into a clear picture of how air quality is distributed, which pollutant drives it and where the highest averages appear. It follows the six steps that are widely used for analytics case studies: ask, prepare, process, analyze, share and act (Google, n.d.). Each step has its own section below, so a reader can follow the work from the question to the decision it supports.

The prepared dataset contains 23,463 records. Excel was used to clean and audit the data and to calculate statistics, and Power BI was used to build an interactive dashboard. The overall Air Quality Index (AQI) has a mean of 72.01 and a median of 55, and the distribution is strongly skewed to the right. Good and Moderate account for about 81.7% of records, but the other 18.3% carry health warnings, and PM2.5 is the pollutant whose average sits closest to the overall index.

Table 1. The project at a glance

Step	Where it is covered
Ask	Section 1: the problem and the business questions
Prepare	Section 2: dataset information and the AQI scale
Process	Section 3: the cleaning process and data audit
Analyze	Section 4: distribution, statistics, categories, pollutants and countries
Share	Section 5: the Power BI dashboard
Act	Section 6: key business insights and decisions they support

Table 2. Headline results

Measure	Result
Prepared records	23,463
Average overall AQI	72.01
Median overall AQI	55
AQI range	6 to 500
Most frequent category	Good (9,936 records)
Records at Unhealthy for Sensitive Groups or worse	4,296 (18.3%)
Highest pollutant average	PM2.5 (68.52)

1. Problem Statement

1.1 The Problem

A dataset with more than twenty thousand air quality records is of little use if it can only be read one row at a time. A decision maker who wants to know how serious air pollution is, which pollutant matters most or which countries stand out needs the data to be cleaned, summarised and shown in a form that can be understood in a few minutes. Without that work, patterns are missed, averages are misread and results are described inconsistently.

1.2 Business Questions

The analysis was organised around four questions that an organisation monitoring or acting on air quality, such as an environmental agency, a public health planner or a company building air quality products, would reasonably ask.

- ✓ How good or poor is air quality across the records, and how are the readings distributed?
- ✓ Which pollutant has the greatest influence on the overall AQI?
- ✓ Which countries show the highest average AQI, and how far can those averages be trusted?
- ✓ How should the results be presented so that decision makers can explore them without technical help?

1.3 Objectives

The objectives were to prepare the data and document its quality, describe the distribution of AQI values and categories, calculate statistics for the overall AQI and pollutant fields, compare country averages, build an interactive dashboard and turn the findings into insights that support decisions. The work is descriptive. It does not explain what caused any reading, forecast future air quality or judge the performance of any country.

2. Dataset Information

2.1 Source and Content

The data comes from the Global Air Pollution Dataset published on Kaggle by hasibalmuzdadid. The prepared workbook contains 23,463 records. It describes the supplied records only and is not a complete measurement of air quality in every country.

Table 3. Fields used in the analysis (country, overall AQI, and the four pollutant AQI fields)

Field	Use in this project
Country	Country comparison; missing values were labelled Unknown
AQI Value (overall)	Statistics, histogram, country averages and the average AQI card
AQI Category (overall)	PivotTable counts, category charts and the dashboard slicer
PM2.5, Ozone, NO2 and CO AQI values and categories	Pollutant comparison and the pollutant visual

2.2 Understanding the AQI

The Air Quality Index runs from 0 to 500, and a higher value means more pollution and a greater health concern. In the United States scale published by the Environmental Protection Agency (EPA), a value of 100 generally corresponds to the national air quality standard for the pollutant, and the scale is divided into six colour coded categories (EPA, n.d.).

Table 4. The six AQI categories

AQI range	Category	Colour	General meaning
0 to 50	Good	Green	Satisfactory; little or no risk.
51 to 100	Moderate	Yellow	Acceptable; unusually sensitive people may be affected.
101 to 150	Unhealthy for Sensitive Groups	Orange	Sensitive groups may experience health effects.
151 to 200	Unhealthy	Red	Everyone may begin to experience health effects.
201 to 300	Very Unhealthy	Purple	Health alert; greater risk for everyone.
301 to 500	Hazardous	Maroon	Emergency conditions; the whole population is likely affected.

An AQI value is worked out separately for each pollutant, and the highest of these becomes the overall AQI (EPA, n.d.). The overall AQI is never lower than any pollutant value in the same record, so a pollutant whose average is close to the overall average is usually the one setting the index.

3. Data Cleaning Process

Cleaning was done in Excel and recorded in two worksheets, Cleaned_Data and Data_Audit. Keeping the audit sheet means that every decision can be traced and the work repeated.

Table 5. Cleaning steps, reasons and outcomes

Step	What was done and why	Outcome
Review structure	Checked the field headings and layout so that each column could be used correctly.	Fields confirmed for analysis
Missing country values	Recorded missing country entries as Unknown instead of deleting the rows, so that the records stay in the overall statistics.	Records kept and clearly marked
Duplicate check	Checked for repeated records that would count the same reading twice.	No duplicates reported
Audit record	Kept a Data_Audit worksheet listing each check.	Work can be repeated
Country comparison filter	Left Unknown rows out of the country averages only, because they cannot be attributed to a country.	Overall figures unaffected

The country filter deserves a clear note. Excluding Unknown records from the country comparison does not remove them from the dataset. They remain in the overall statistics and in the category counts, and the totals in this study use all 23,463 prepared records.

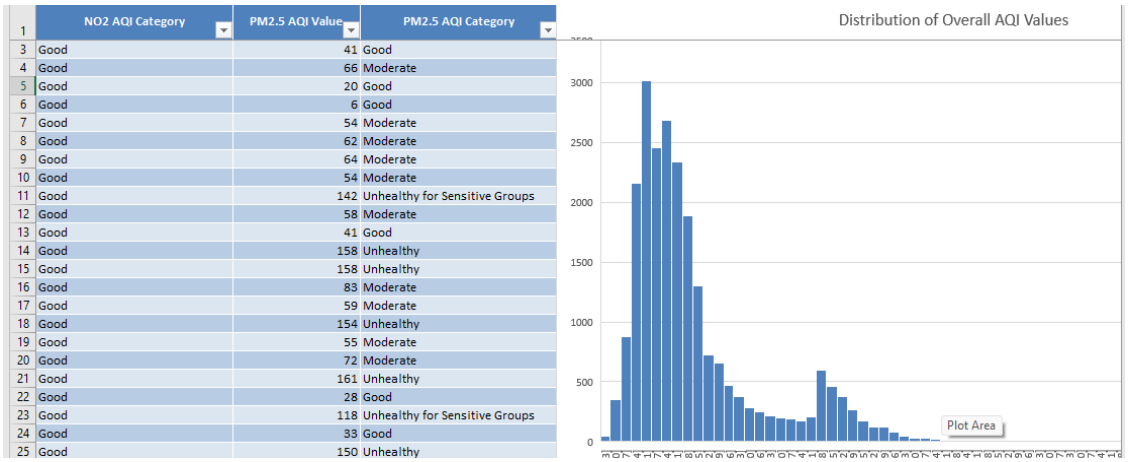


Figure 1. The Cleaned_Data worksheet and the histogram of overall AQI values in Microsoft Excel.

4. Analysis

4.1 Distribution of Overall AQI

The histogram in Figure 1 shows most observations in the lower AQI ranges and a thinner tail stretching toward higher values, with a small secondary rise further along the axis. The long right tail matches the positive skewness found in the statistics.

4.2 Descriptive Statistics

The mean of 72.01 is higher than the median of 55, a gap of about 17 points. This is the typical signature of a right skewed distribution, where a relatively small number of high readings pull the average up while the median stays close to the typical record. Both values fall in the Moderate band, so the summary is stable at category level, but the median describes the typical record better.

Table 6. Descriptive statistics for overall AQI

Statistic	Overall AQI
Mean	72.01
Median	55
Mode	50
Minimum	6
Maximum	500
Skewness	Approximately 3.29

4.3 AQI Categories

A PivotTable counted the records in each category. Good was most frequent with 9,936 records (42.3%), followed by Moderate with 9,231 (39.3%), and together they hold 19,167 records or about 81.7%. Records at Unhealthy for Sensitive Groups or worse total 4,296 (18.3%), and records at Unhealthy or worse total 2,705 (11.5%). The percentages were rechecked against the counts and the total of 23,463.

Table 7. Records by AQI category

AQI category	Records	Share
Good	9,936	42.3%
Moderate	9,231	39.3%
Unhealthy for Sensitive Groups	1,591	6.8%
Unhealthy	2,227	9.5%
Very Unhealthy	287	1.2%
Hazardous	191	0.8%
Total	23,463	100%

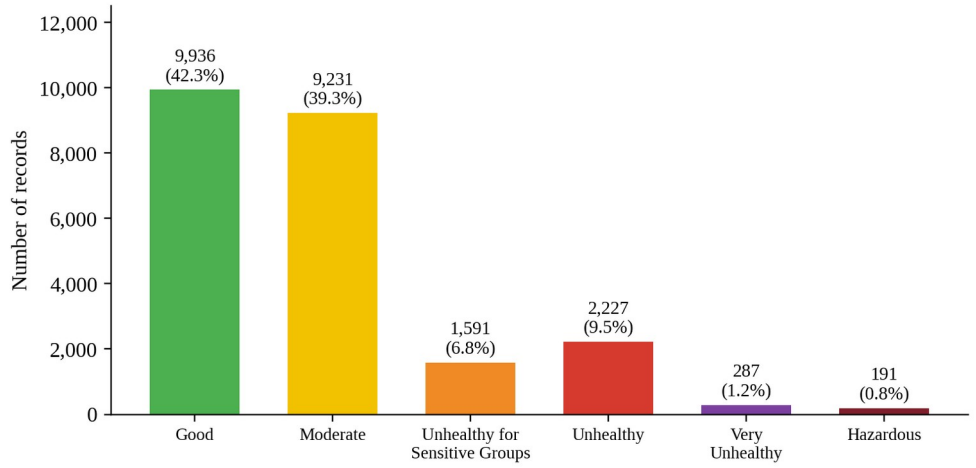


Figure 2. Records by AQI category, redrawn from Table 7 using the standard AQI colours.

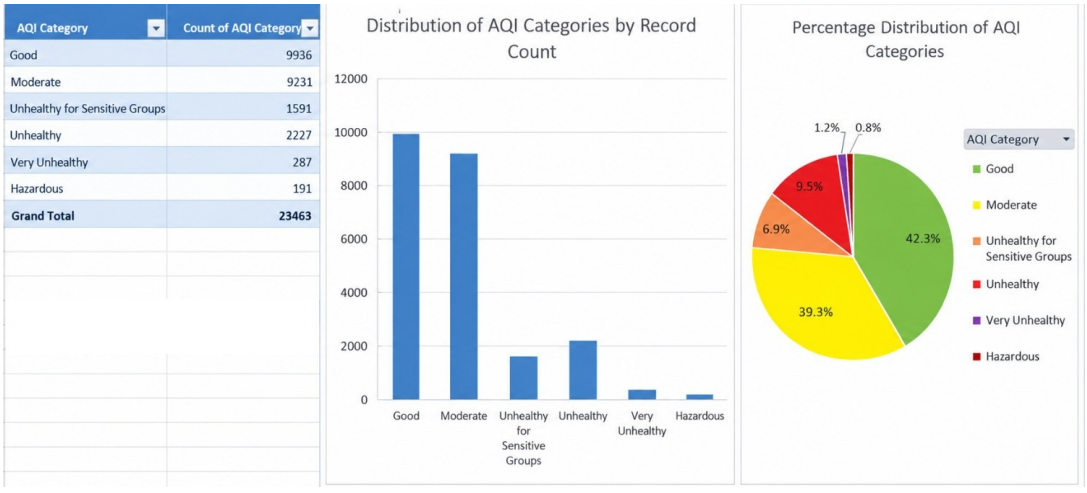


Figure 3. The AQI category PivotTable with the record count chart and the percentage chart in Excel.

4.4 Pollutants

PM2.5 had the highest average among the pollutant specific measures, at 68.52. Because the overall AQI takes the highest pollutant value in each record, a PM2.5 mean of 68.52 sitting

close to an overall mean of 72.01 suggests that PM2.5 is the pollutant that most often sets the overall value in this dataset. This points to PM2.5 as the leading pollutant in the dataset.

4.5 Country Comparison

The country worksheet lists the ten highest average AQI values after Unknown records were excluded. The Republic of Korea is highest at 421.00, followed by Bahrain (188.00), Mauritania (179.00), Pakistan (178.79) and the United Arab Emirates (163.67).

Table 8. The ten highest country averages of overall AQI

Rank	Country	Average AQI
1	Republic of Korea	421.00
2	Bahrain	188.00
3	Mauritania	179.00
4	Pakistan	178.79
5	United Arab Emirates	163.67
6	Aruba	163.00
7	Kuwait	162.00
8	Qatar	157.50
9	India	152.96
10	Senegal	152.42

The Republic of Korea stands more than twice as high as the next country. Country averages depend on how many observations each country contributes and over what period, so they describe the records in this dataset and are best read alongside record counts. The table is not a ranking of national air quality.

5. Dashboard

5.1 Design

The Power BI dashboard, titled Global Air Quality Dashboard, answers the fourth business question. It keeps the headline numbers on the left, the distributions in the middle and the map at the bottom, so a viewer moves from the overall picture to the detail without searching. A slicer at the top filters the whole page by AQI category.

Table 9. Dashboard components

Component	Purpose
Average AQI card	Shows the overall average AQI of 72.01.
Total AQI Records card	Shows the size of the prepared dataset, 23,463 records.
AQI Category Distribution	Compares record counts across the six categories.
Average AQI Pollutant	Shows pollutant related averages in a doughnut visual.
Average AQI by Country	Gives a map view of country level averages.
AQI Category slicer	Filters every visual by the chosen category.

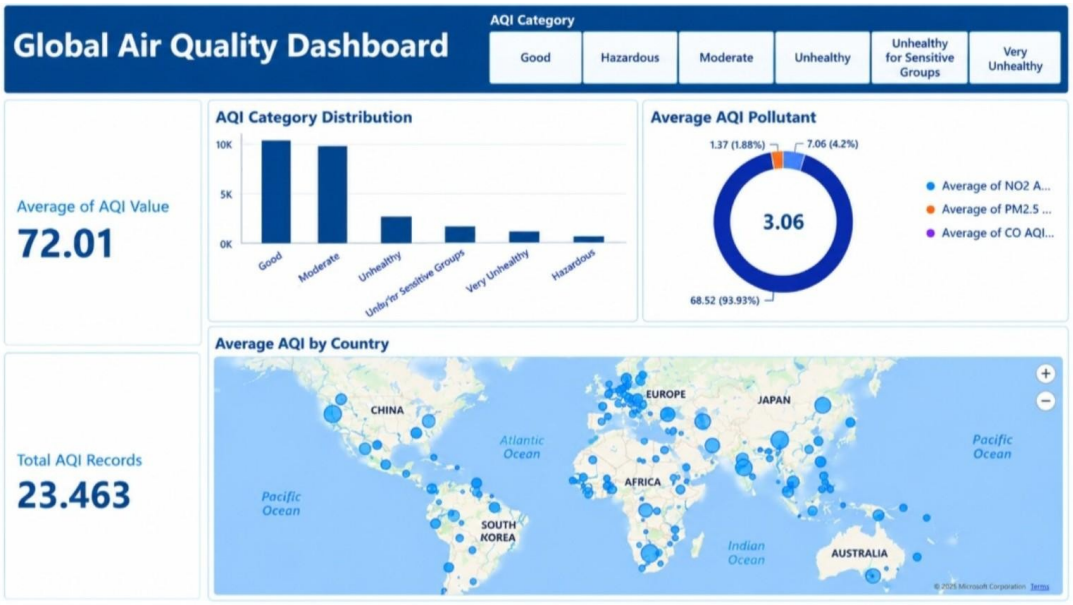


Figure 4. The Global Air Quality Dashboard developed in Microsoft Power BI.

5.2 Reading the Dashboard

The cards give the headline measures, the category chart shows how records spread across the six bands, and the pollutant visual shows PM2.5 as by far the largest average. The map adds geographic context, and the slicer lets a viewer isolate, for example, only the Hazardous records to see where they appear. The dashboard is an overview for exploration and does not replace the statistics in Section 4.

6. Key Business Insights

This section turns the findings into decisions. Each insight states the evidence and the decision it supports, and the section closes with the three recommendations that follow from them.

Table 10. Insights, evidence and the decisions they support

Insight	Evidence	Decision it supports
Most records show acceptable air	Good and Moderate make up 81.7% of the 23,463 records.	Use these categories as the baseline when judging whether a location or period is unusual.
Attention belongs on the tail, not the average	Mean 72.01 sounds Moderate, yet 18.3% of records carry health warnings and 191 are Hazardous.	Set alerts and monitoring priorities on the share of high AQI records.
One average does not tell the whole story	The mean (72.01) is 17 points above the median (55); skewness is about 3.29.	Report the median and category shares next to the mean in every summary.
Particulate matter drives the index	PM2.5 averages 68.52, close to the overall mean of 72.01.	Direct monitoring effort and mitigation planning toward fine particles first.
Country averages are best read with context	Korea averages 421.00, more than twice the next country (Bahrain, 188.00).	Present country averages together with record counts and the period covered.

6.1 Top Three Recommendations

- Monitor the high end of the scale. Base alerts and reporting on the share of records at Unhealthy for Sensitive Groups or worse (18.3%), supported by the median and the category counts, instead of relying on the average alone.
- Put fine particles first. Because PM2.5 sits closest to the overall AQI, give PM2.5 monitoring and reduction the highest priority when resources are limited.
- Use the dashboard as the standard view. Share the Global Air Quality Dashboard with decision makers, and show record counts and time coverage beside any country comparison so that results are read fairly.

7. Scope and Limitations

The scope of the project is set by the data it uses. The points below describe what the results cover, so that they are used in the right way.

- ✓ The analysis covers the records and fields in the selected Kaggle dataset and describes the patterns across them.
- ✓ Country figures are averages of the records supplied for each country, so they are read together with record counts and time coverage and are not a ranking of national air quality.
- ✓ The work is descriptive. It shows how AQI is distributed and which pollutant leads, and it does not assign causes to individual readings.
- ✓ The AQI is a communication scale tied to national standards. The World Health Organization (2021) publishes separate and generally stricter guidelines, so a Good reading does not mean there is no health risk.

7.1 Next Steps

The study can be extended in three directions. Adding dates would allow seasonal patterns and trends to be studied. Adding record counts to the country map would make country comparisons easier to read. Comparing the category boundaries with the current EPA breakpoints, which were revised for PM2.5 in 2024, would bring the results in line with the latest guidance.

8. Conclusion

This project took a global air pollution dataset of 23,463 records from a raw spreadsheet to a cleaned workbook, a set of statistics and an interactive dashboard. Most records are Good or Moderate, but a sizeable minority carry health warnings, the distribution is strongly skewed, and PM2.5 is the pollutant closest to the overall index. The country comparison put the Republic of Korea far ahead, but that result depends on record counts and coverage and should not be read as a ranking.

The main lesson for anyone using this kind of data is to report more than one number, to show how many records lie behind each figure and to keep a record of every cleaning decision. With those habits, a large environmental dataset can be turned into insights that decision makers can understand and act on.

References

Google. (n.d.). Google Data Analytics Capstone: Complete a case study. Coursera. https://www.coursera.org/learn/google-data-analytics-capstone

hasibalmuzdadid. (n.d.). Global Air Pollution Dataset. Kaggle. https://www.kaggle.com/datasets/hasibalmuzdadid/global-air-pollution-dataset

Microsoft. (n.d.). Power BI documentation. https://learn.microsoft.com/power-bi/

U.S. Environmental Protection Agency. (n.d.). Air Quality Index (AQI) basics. AirNow. https://www.airnow.gov/aqi/aqi-basics/

World Health Organization. (2021). WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. https://www.who.int/publications/i/item/9789240034228